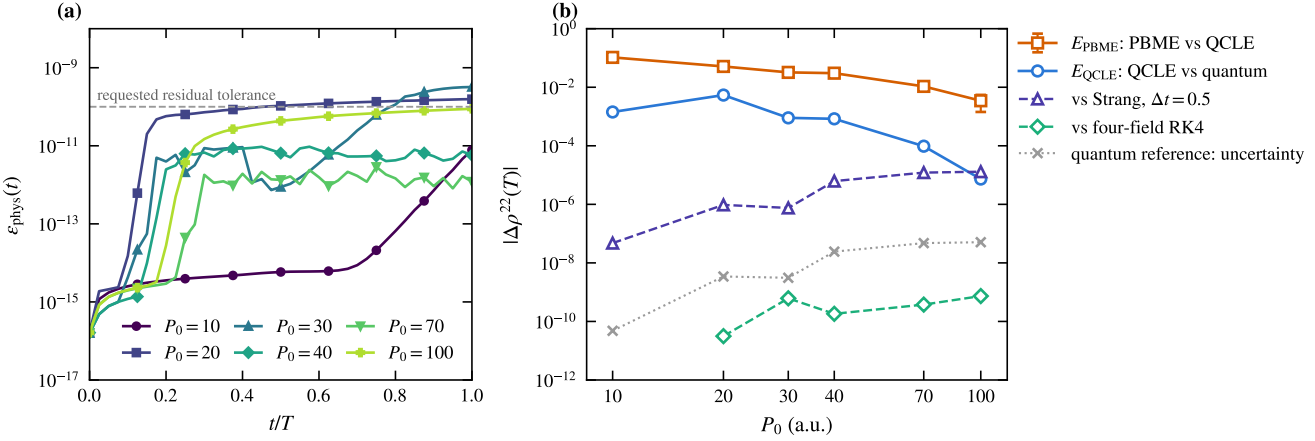
(a)
requested residual tolerance
$\epsilon_{\mathrm{phys}}(t)$
t/T
$P_0 = 10$
$P_0 = 20$
$P_0 = 30$
$P_0 = 40$
$P_0 = 70$
$P_0 = 100$
(b)
$|\Delta\rho^{22}(T)|$
P_0 (a.u.)
E_{PBME}: PBME vs QCLE
E_{QCLE}: QCLE vs quantum
vs Strang, $\Delta t = 0.5$
vs four-field RK4
quantum reference: uncertainty